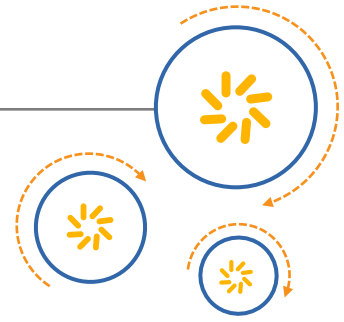




Qualcomm Technologies, Inc.



WGR7640 IC

Device Revision Guide

September 2016

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LM80-P0436-30 Rev. B

Revision history

Revision	Date	Description
B	September 2016	Updated "E" part
A	July 30, 2015	Initial release

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1 Introduction

This document contains a description of the chipset capabilities. Not all features are available, nor are all features supported in the software.

NOTE: Enabling some features may require additional licensing fees.

This device revision guide describes the WGR7640, a GNSS receiver IC.

Technical information is primarily covered by the documents listed in [Table 1-1](#). Each is self-contained, but a thorough understanding of the device and its applications requires familiarization with it.

Table 1-1 Primary WGR documents

Document number	Document title
LM80-P0436-31	<i>WGR7640 GNSS RF Receiver IC Device Specification</i>

1.1 Scope and intended audience

This device revision guide identifies issues with all WGR7640 samples released to date. The following information is included:

- Introduction to this document and its topic (this chapter):
- Device identification (Chapter [2](#))
 - ❑ Device marking
 - ❑ Hardware revision number
 - ❑ Identification details for each sample type
 - ❑ Sample testing – explanations of engineering samples (ES) and commercial samples (CS)
- Known issues (Chapter [3](#))
 - ❑ Issue description
 - ❑ Impact to system performance
 - ❑ Possible workarounds (what designers should do to minimize the issue’s impact)

This device revision guide is intended for new product developers who are designing, testing, and/or evaluating phones or terminals that include the WGR7640 device.

1.2 Acronyms, abbreviations, and terms

Table 1-2 provides definitions for the acronyms, abbreviations, and terms used in this document.

Table 1-2 Acronyms, abbreviations, and terms

Term	Definition
CS	Commercial samples
ES	Engineering samples
WGR	WLP GPS receiver

1.3 Additional information

For additional information, go to

<https://developer.qualcomm.com/hardware/snapdragon-410/tools>

2 Device Identification

The WGR7640 device can be identified by markings on its top surface and by the contents of an identification register; these identification techniques are described in Section 2.1 and Section 2.2. Further details about each sample type are presented in Section 2.3 and Section 2.4.

2.1 Device marking – general format

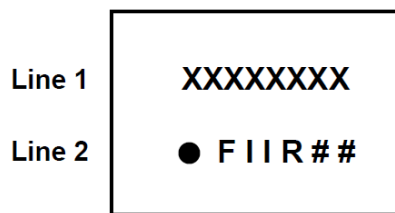


Figure 2-1 WGR7640 part marking (top view – not to scale)

Table 2-1 Part marking line description

Line	Marking	Description
1	XXXXXXXX	XXXXXXXX = traceability number
2	FIIR##	F = supply source code - F = A for TSMC II = item identifier - See Table 2-2 for assigned values R = product revision – also indicates hardware ID value - See Table 2-2 for assigned values ## = 2 digit wafer number

Table 2-2 Device identification code details

WGR variant	II value	R value
ES sample type		
WGR7640	03	1
CS sample type		
WGR7640	03	2

NOTE: For complete marking definitions of all WGR7640 IC variants and revisions, refer to the *WGR7640 GNSS RF Receiver IC Device Specification* (LM80-P0436-31).

2.2 Hardware revision number – general format

The device identification register allows the user to determine the device's manufacturer, part number, and version through the test access port. The 32-bit device identification register is read through the JTAG interface and is summarized in [Table 2-3](#).

Table 2-3 Device identification register

Bit location	Description	Value
SSBI 0X9D<0>	ES	0
SSBI 0X9D<0>	CS	1

2.3 Device identification for each sample type

The device can be identified by the code, as illustrated in [Figure 2-2](#).

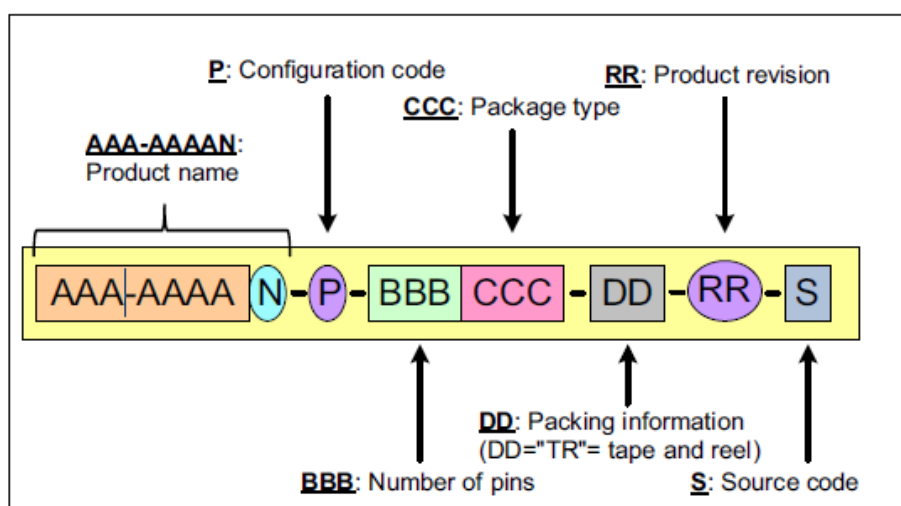


Figure 2-2 Device identification code

The product revision code (R) reflects only die revisions. A source configuration code (S) has been added to reflect all qualified sourcing combinations (i.e., multiple F codes).

An example can be as follows: WGR-7640-0-17WLN-SP-TR-00-0

2.4 Sample testing

2.4.1 Engineering samples

These devices have undergone limited testing and sometimes have significant feature limitations. They are suitable to assist with printed circuit board development, to conduct board-level electrical evaluation tests, and to explore manufacturing considerations. Engineering samples must not be used for phone-level qualification.

2.4.2 Commercial samples

These devices have undergone full production-level testing and meet the specifications and features described in the device specification, except as otherwise noted in this document. They have passed device-level qualification. Commercial samples (CS) are suitable for use in performance testing and phone-level production and qualification.

3 Known Issues

3.1 Summary of known issues

All known issues for each revision of the WGR7640 device are summarized in [Table 3-1](#). The text within the **Issue** column refer to the sections of this document that explain the issues, regardless of the sample type (or types) on which they occur. An *X* in any of the other columns indicates that the issue occurs on the corresponding sample type.

Table 3-1 Known issues – All sample types and revisions

No.	Issue	WGR7640 ES	WGR7640 CS
		II = 03	II = 03
		R = 1	R = 2
1	Table 3-2 Issue 1 – RSB failure at cold temperature	X	X
2	Table 3-3 Issue 2 – Beidou mode performance is not confirmed at platform level	X	X
3	Table 3-4 Issue 3 – Gain compression failure with high jammer	X	

3.1.1 Issues – description, impact, and workaround

Table 3-2 Issue 1 – RSB failure at cold temperature

Description	RSB failure at cold temperature
Impact	Low C/N performance
Workaround	Ensure that the 1.3 V VDD_RF voltage at the WGR7640 input does not drop below 1.2 V.

Return to [Table 3-1](#)

Table 3-3 Issue 2 – Beidou mode performance is not confirmed at platform level

Description	Beidou mode performance is not confirmed at the platform level.
Impact	Since the Beidou digital engine is not available at the time for CS, this does not impact anything at the platform level.
Workaround	When the Beidou digital engine is CS, the platform performance will be verified.

Return to [Table 3-1](#)

Table 3-4 Issue 3 – Gain compression failure with high jammer

Description	Gain compression fails with a high jammer.
Impact	C/N is affected with a higher jammer.
Workaround	This issue is fixed in CS.

Return to [Table 3-1](#)

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